# Agenda

Example

Polynomial Regression

Piecewise Polynomials

Cubic splines

$Y=1$ Dough rose

$Y=2$ Dough didn't rise

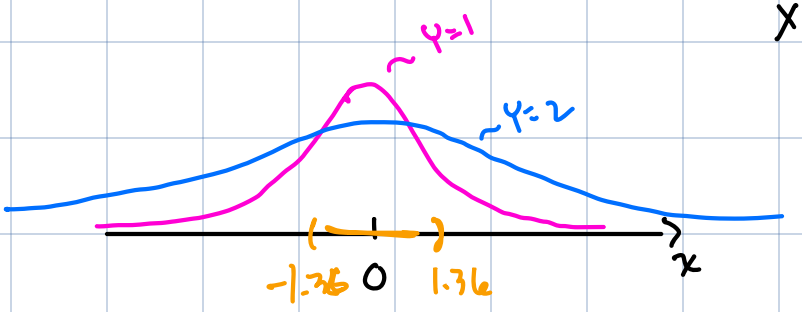
$X$: proofing time (centered)

$P(Y=1) = .5 = \hat{\pi}_1$

$P(Y=2) = .5 = \hat{\pi}_2$

$X|Y=1 \sim N(0,1)$

$X|Y=2 \sim N(0,4)$




What values of $x$ will we predict $Y=1$?

$$\delta_1(x) = \log\left(\frac{1}{2} \times \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)\right) = \log\frac{1}{2} - \frac{1}{2}\log 2\pi - \frac{x^2}{2}$$

$$\delta_2(x) = \log\left(\frac{1}{2} \times \frac{1}{\sqrt{2\pi}2} \exp\left(-\frac{x^2}{8}\right)\right) = \log\frac{1}{2} - \frac{1}{2}\log 2\pi - \log 2 - \frac{x^2}{8}$$

$\delta_1(x) > \delta_2(x)$

$$-\frac{x^2}{2} > -\log 2 - \frac{x^2}{8}$$

$\pi_k f_k(x)$

$\vdots$

$\Rightarrow |x| < 1.36$

# 1. Polynomial Regression

## 1.1 Model

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$

$$d.f. = d + 1 \quad (\#\text{ of coefficients})$$

d.f. : # of knobs the fit can turn to specify the curve. Higher means more flexible.

## 1.2 Distn of $\hat{y}$

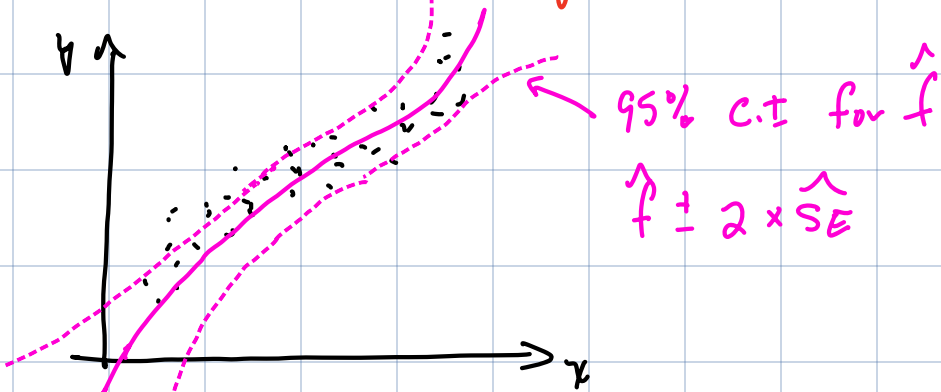
$$\hat{y} \sim N(\mathbb{B}\beta, \sigma^2 \mathbb{H}) \qquad \mathbb{H} = \mathbb{B}(\mathbb{B}^T\mathbb{B})^{-1}\mathbb{B}^T$$

$$Var(\hat{f}(x_i)) = \sigma^2 h_{ii}$$

Sum this over entire training set

$$\sum_{i=1}^{n} Var(\hat{f}(x_i)) = \sigma^2 \, tr(\mathbb{H}) = \sigma^2 (d+1) = \sigma^2 \, d.f.$$

Every choice of basis gives same total variance for a fixed d.f., but changes where the variance is.



We have $x_0$, $b(x_0) = \begin{pmatrix} 1 \\ x_0 \\ x_0^2 \\ \vdots \\ x_0^d \end{pmatrix}$

$$\Rightarrow \hat{f}(x_0) = b(x_0)^T \hat{\beta}$$

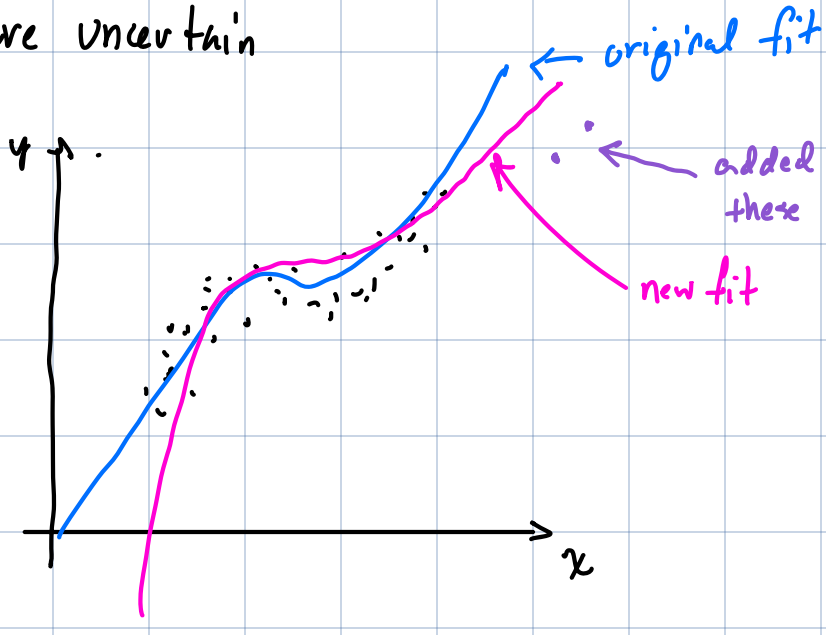
$$Var(\hat{f}(x_0)) = Var(b(x_0)^T \hat{\beta}) = b(x_0)^T Var(\hat{\beta})\, b(x_0)$$

$$= \sigma^2 b(x_0)^T (B^T B)^{-1} b(x_0)$$

$$\hat{\sigma}^2 = \frac{RSS}{n-d-1} \Rightarrow \hat{SE}(\hat{f}(x_0)) = \sqrt{\hat{Var}(\hat{f}(x_0))}$$

1.3 Failure points

- Edge variance is high
- Tails are uncertain




Why? the curve uses ALL of the data.
It's fit "globally"

## 1.4 Choosing $d$

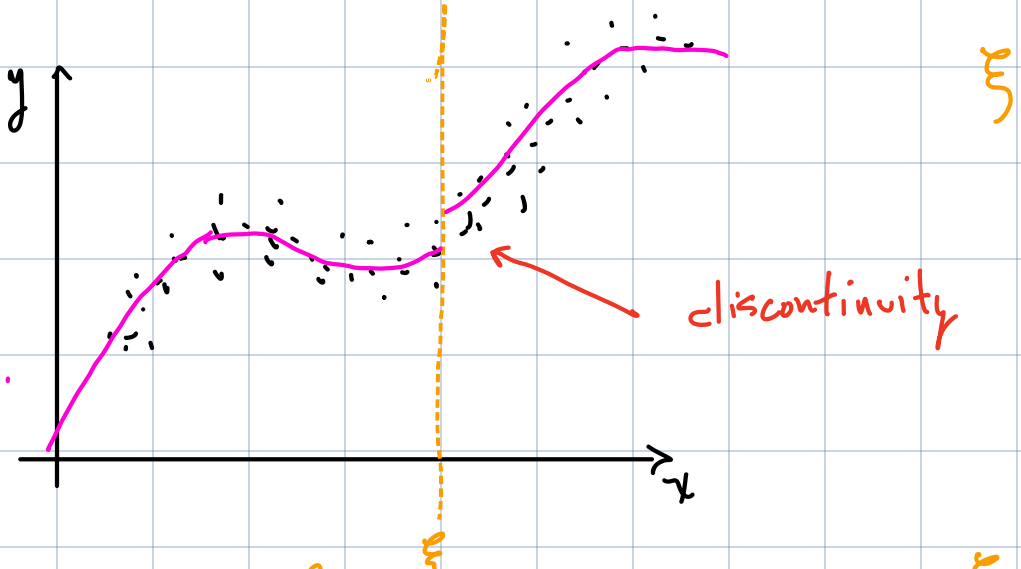
- Use cross-validation
- F-tests

Works on logistic reg. too!

$$P(y_i = 1 \mid x_i) = \frac{1}{1 + \exp\left(-(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)\right)}$$

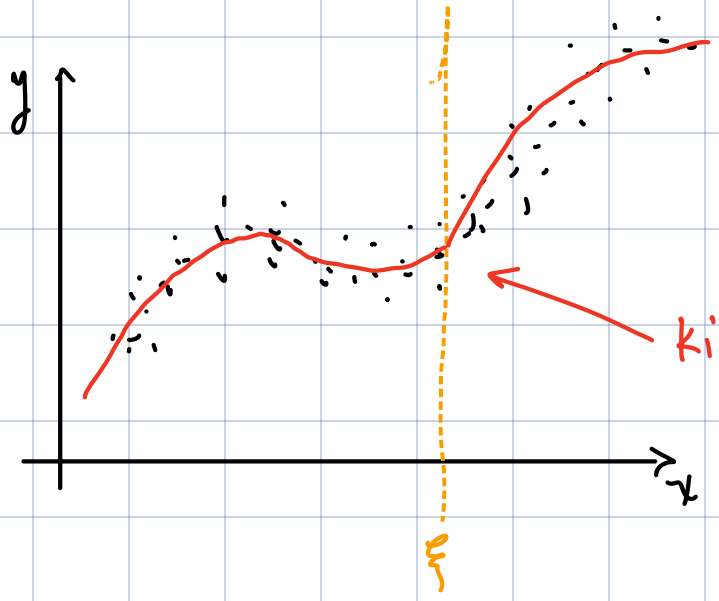
## 2. Piecewise Polynomials and Regression Splines

### 2.1 Knots

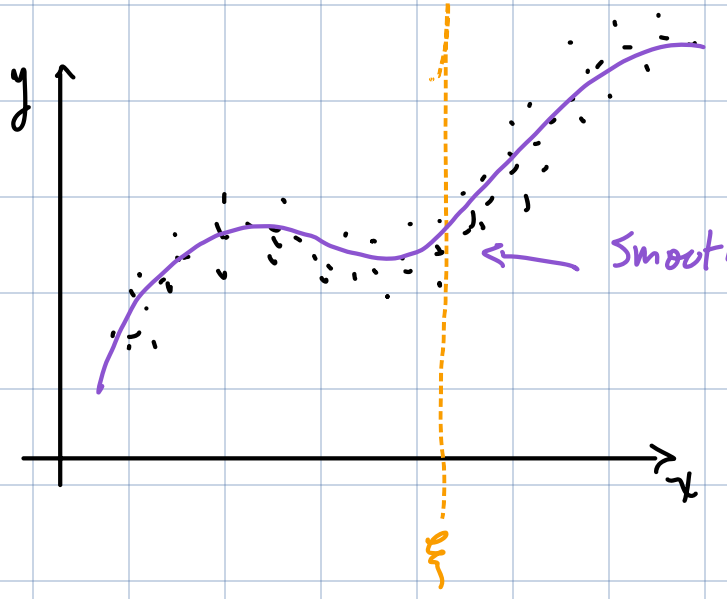



$$y_i = \begin{cases} \beta_{01} + \beta_{11} x_i + \beta_{21} x_i^2 + \beta_{31} x_i^3 + \varepsilon_i & x_i < \xi \\ \beta_{02} + \beta_{12} x_i + \beta_{22} x_i^2 + \beta_{32} x_i^3 + \varepsilon_i & x_i \geq \xi \end{cases}$$

8 parameters (8 d.f.)




d.f = 7

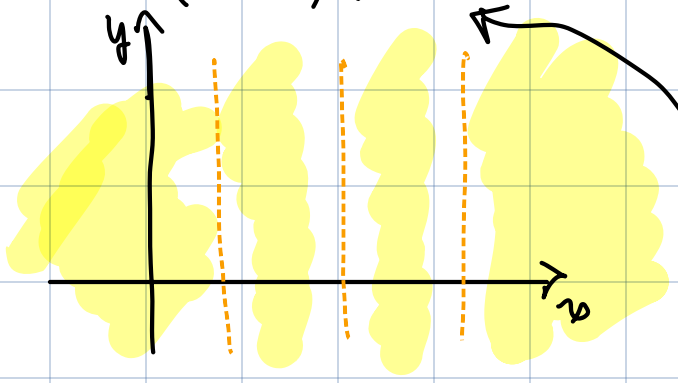



d.f 6

| Knot constraints | d.f. | Name |
|---|---|---|
| none | 8 | piecewise cubic |
| $f$ continuous | 7 | continuous piecewise |
| $f, f'$ continuous | 6 | — |
| $f, f', f''$ continuous | 5 | cubic spline |
| $f, f', f'', f'''$ continuous | 4 | no knot |

In general, a degree $d$ spline with knots $\xi_1, \dots, \xi_K$, has

$$\text{d.f.} = (K+1)(d+1) - Kd = K + d + 1$$




$K+1$ polynomials

# of coefficients

$d$ constraints per knot

Cubic: $d = 3$

d.f.: $K+4$

total d.f w/out constraints

## 2.2 Cubic splines (d.f. $K+4$)

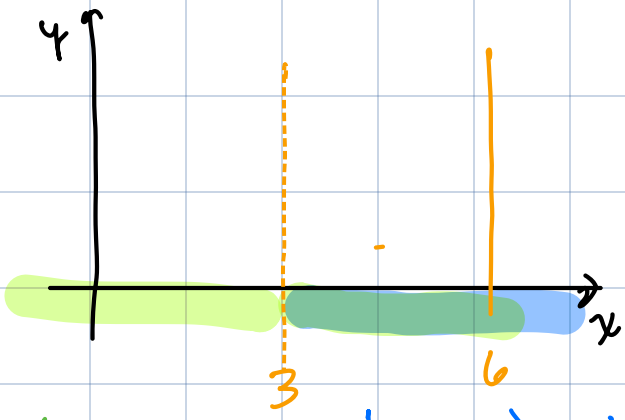
$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \beta_3 b_3(x_i) + \cdots + \beta_{K+3} b_{K+3}(x_i) + \varepsilon_i$$

Ex. 3 knots gives us $\beta_4, \beta_5, \beta_6$

What are $b_4, b_5, b_6$?

Use truncated power basis!

$$h(x, \xi) = (x-\xi)_+^3 = \begin{cases} (x-\xi)^3 & x > \xi \\ 0 & \text{else} \end{cases}$$




$h(x,6) = (x-6)^3 \quad x \geq 6$

$0 \quad x < 6$

$h(x,3) = 0$

$h(x,3) = (x-3)^3$